

Week 8 Case Study

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INFT1203: Access Controls

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Best Practices

Avoid Unnecessary Software

Implementation

The Ubuntu Server Security Practice “Avoid Unnecessary Software” will be implemented in our medical drone logistics organization with a continuous process that is highly relevant to our drone delivery operations and threat modeling. As our organization grows and changes, our systems and threat landscape will change which means our collection of critical tools and software will also change. For example, developers may install temporary tools for debugging which must be removed once that tool is no longer used to reduce the attack surface of the organization.

Initially, it is important to be able to identify the roles of each used Ubuntu server and the allowed software per server. Whether an Ubuntu server is used for flight operations control, telemetry ingestion, or a workstation for developers, we will classify the intended use of all servers within the organization. From there, we will create an approved software list for each role, which will specify the allowed packages, snaps, and services. Only strictly required software will be provided for the role of each server, and anything else will be implicitly forbidden.

Our organization will use minimal installation versions for our Ubuntu servers and not include all optional server snaps during installation. OpenSSH or other required roles will only be installed. Images will be developed that will install a pre-approved set of packages per identified role for specific Ubuntu servers.

Our organization will have separate developer and production environments to further reduce compromise capabilities. Our developers and data scientists will have their own sandboxes or virtual machines such that their production tools such as compilers, debuggers, and packet sniffers are not found in our production servers.

Furthermore, the command “apt” and similar commands will be restricted on production and developer servers. Tickets or approvals will be required to add a new package to a server and logs will be stored for all packages that are installed.

Importance

Ubuntu Linux Operating Systems typically contain packages that cover many use cases where a production server is unlikely to use the same packages. Although, a production server will have critical workloads to perform where many packages of a typical Ubuntu server should be removed as they will be unused. Therefore, not all services on an Ubuntu server are necessary and unnecessary services should be removed to reduce the attack surface of the server.

With all modes of implementation used in our company, our Ubuntu servers will stay small and simple while behaving in a way that staff and management can reliably understand and control these servers. Additionally, our Ubuntu servers will have minimal attack surface, resource usage, configuration complexity which keeps server maintenance over time. Finally, our servers will be predictable, associating roles with software running on those servers. This will make it easy to identify and detect software anomalies and misconfigurations that increases reliability in a medical logistics environment.

[Best Practice 2]*Implementation**Importance*

[Best Practice 3]*Implementation**Importance*

References

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